

Note 13.3 Arc Length in Space

Arc Length Along a Space Curve

Definition of the Length

The **length** of a smooth curve $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, $a \leq t \leq b$, that is traced exactly once as t increases from $t = a$ to $t = b$, is

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

That is,

$$L = \int_a^b |\mathbf{v}| dt$$

where $|\mathbf{v}|$ is the length of a velocity vector $d\mathbf{r}/dt$.

Arc Length Parameter with Base Point $P(t_0)$

If we choose a base point $P(t_0)$ on a smooth curve C parametrized by t , each value of t determines a point $P(t) = (x(t), y(t), z(t))$ on C and the **arc length parameter** for the curve is

$$s(t) = \int_{t_0}^t |\mathbf{v}(c)| dc = \int_{t_0}^t \sqrt{[x'(c)]^2 + [y'(c)]^2 + [z'(c)]^2} dc$$

That is to use the as parameter distance s rather the time t to express the position function $\mathbf{r}$.

Speed on a Smooth Curve

The derivative of s with respect to t is the speed of the moving particle (by Fundamental Theorem of Calculus). That is,

$$\frac{ds}{dt} = |\mathbf{v}(t)|$$

Unit Tangent Vector

The **unit tangent vector** is

$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|}$$

or for the differentiable vector function $\mathbf{r}$, by the chain rule we have

$$\frac{d\mathbf{r}}{ds} = \frac{d\mathbf{r}}{dt} \frac{dt}{ds} = \mathbf{v} \frac{1}{|\mathbf{v}|} = \mathbf{T}$$

That is the information of direction of the velocity.

Exercises

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